

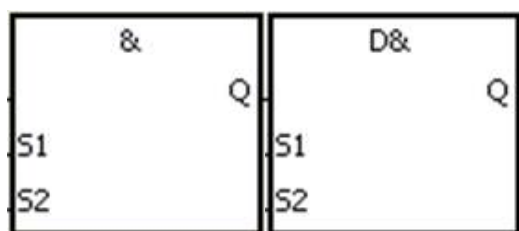
API	Instruction code			Operand	Function
0809- 0811	D	LD #		S_1, S_2	Contact type of logical operation LD #

Device	X	Y	M	S	T	C	HC	D	FR	SM	SR	E	K	16#	"\$"	F
S_1	●	●			●	●	●	●	●		○	○	○	○		
S_2	●	●			●	●	●	●	●		○	○	○	○		

Data type	BOOL	WORD	DWORD	LWORD	UINT	INT	DINT	LINT	REAL	LREAL	TMR	CNT	STRING
S_1		●	●			●	●				●	●	
S_2		●	●			●	●				●	●	

Pulse instruction	16-bit instruction	32-bit instruction
-	AS	AS

Symbol



S_1 : Data source 1

S_2 : Data source 2

Taking LD& and DLD& for example

Explanation

- The instruction is used to compare the data in S_1 with that in S_2 . When the comparison result is not 0, the condition of the instruction is met. When the comparison result is 0, the condition of the instruction is not met.
- Only the instruction DLD # can use the 32-bit counter but not the device E.
- The instruction LD # can be connected to the mother line directly.

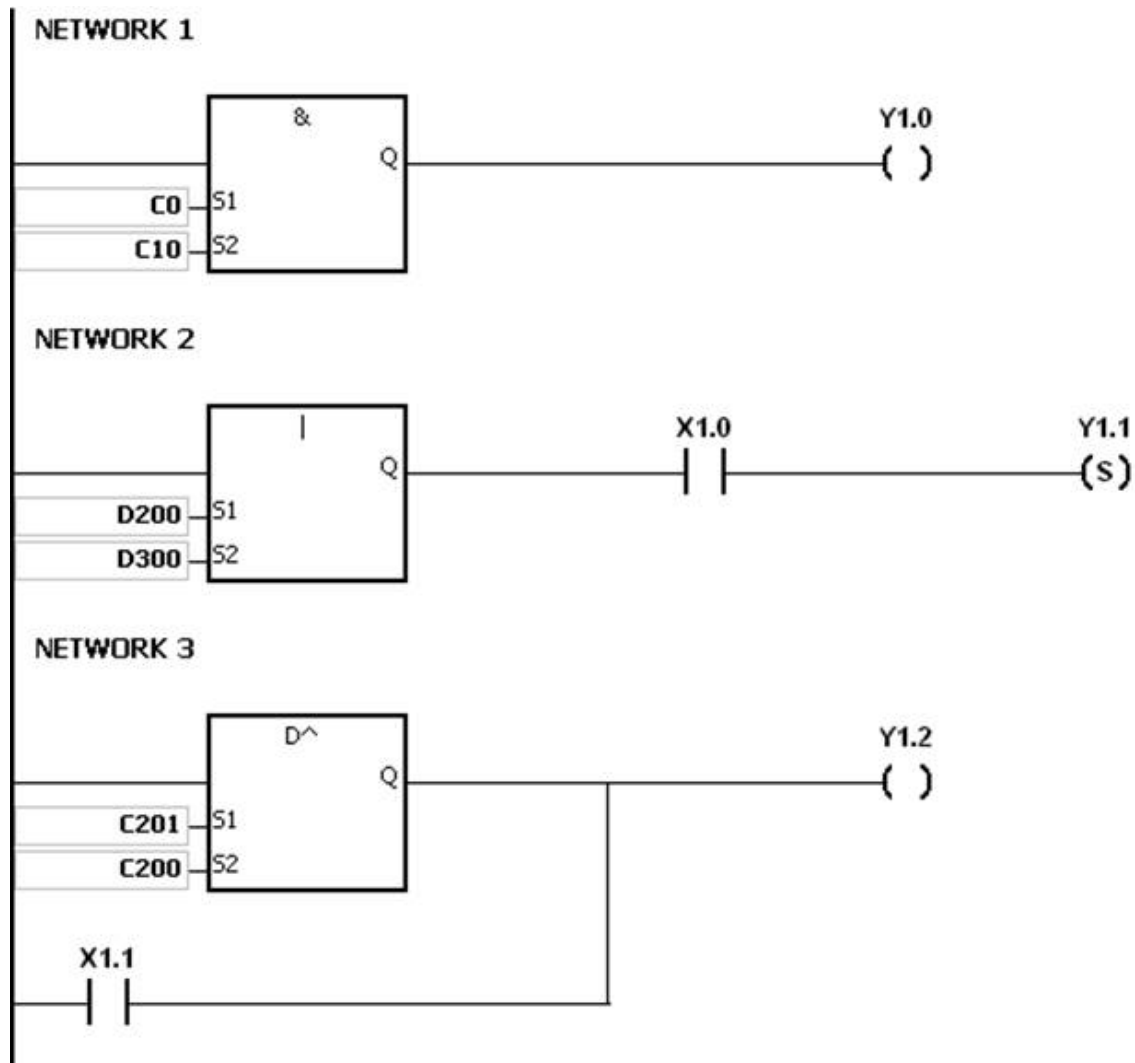
API No.	16-bit instruction	32-bit instruction	Comparison operation result	
			ON	OFF
0809	LD&	DLD&	$S_1 \& S_2 \neq 0$	$S_1 \& S_2 = 0$
0810	LD	DLD	$S_1 S_2 \neq 0$	$S_1 S_2 = 0$
0811	LD^	DLD^		$S_1 \wedge S_2 = 0$

			$s_1 \wedge s_2 \neq 0$	
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4. &: Logical AND operation
5. |: Logical OR operation
6. ^: Logical exclusive OR operation

Example

1. The logical operator AND takes the data in C0 and C1, and performs the logical AND operation on each pair of corresponding bits. When the operation result is not 0, Y1.0 is ON.
2. The instruction performs the logical operation OR on each pair of corresponding bits in D200 and D300, when the operation result is not 0 and X1.0 is ON, Y1.1 is ON.
3. The instruction performs logical exclusive operation XOR on each pair of corresponding bits in C201 and C200, when the operation result is not 0, or when X1.1 is ON, Y1.2 is ON.

**Additional remarks**

If S_1 or S_2 is illegal, the condition of the instruction is not met, SM0 is ON, and the error in SR0 is 16#2003.